
Project Plan

COSC 70: Software Engineering I

VenueVue: Offline Point-of-Sale and Inventory Management System for Mobile Pop-Up Coffee Shops

TRAVELBEAN MNL

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Document Change Control

This section provides control for the development and distribution of revisions to the Project Charter up to the point of approval. The Project Charter does not change throughout the project life cycle, but rather is developed at the beginning of the project (immediately following project initiation approval, and in the earliest stages of project planning). The Project Charter provides an ongoing reference for all project stakeholders. The table below includes the revision number (defined within your Documentation Plan Outline), the date of update/issue, the author responsible for the changes, and a brief description of the context and/or scope of the changes in that revision.

Revision Number	Date of Issue	Author(s)	Brief Description of Change

Disclaimer

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Abstract

Mobile pop-up coffee vendors operate in dynamic physical environments where permanent internet connectivity is absent or prohibitively unstable^[5]. TravelBean MNL, a mobile coffee venture based in Tondo, Manila, faces severe operational bottlenecks due to manual paper-based order taking, post-event calculator sales tallying, and unmonitored raw material consumption. Existing commercial point-of-sale (POS) systems either depend strictly on cloud infrastructure or fail to track inventory at the ingredient level (e.g., deducting coffee beans in grams, milk in milliliters, and paper cups in units per beverage sold)^[1].

VenueVue, engineered by group CTRL+Stay, is an offline-first Point-of-Sale and Inventory Management System engineered specifically to eliminate these operational gaps^[1]. Designed around a local client-server model, VenueVue leverages a laptop running an offline XAMPP environment (PHP 8.x and MySQL 8.x) that generates a local Wi-Fi hotspot. Tablet devices utilized by baristas connect directly to this local network via web browsers, permitting real-time order processing, automatic recipe-based ingredient deduction, and instant low-stock threshold alerting without internet connectivity^[1]. Data reconciliation across devices is handled post-event via local Wi-Fi push, direct USB file transfer, or manual structured Excel/CSV imports^[1].

This project plan details the complete software development lifecycle for VenueVue, encompassing a 5-month (20-week) timeline across planning, analysis, design, development, testing, and deployment. By combining structured Systems Development Life Cycle (SDLC) documentation governance with Agile sprint cycles, group CTRL+Stay delivers a robust, zero-subscription software solution that reduces post-event administrative overhead from hours to minutes while providing granular business analytics via automated four-tab Excel reports^[4].

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The Problem Domain

Statement of the Problem

Mobile pop-up coffee businesses operating in temporary physical venues—such as outdoor fairs, private weddings, school events, and corporate pop-ups—face severe operational inefficiencies due to complete reliance on manual paper-based sales tracking and visual inventory estimation. Because event venues frequently lack reliable cellular coverage or public Wi-Fi networks, cloud-based point-of-sale (POS) systems become unusable^[1]. Consequently, pop-up vendors suffer from order-taking errors, lost sales logs, stockouts of critical ingredients mid-event, and an inability to analyze event profitability^[6].

A quantitative SWOT and problem prioritization analysis conducted by group CTRL+Stay with the client identified seven distinct specific problems affecting daily operations:

Priority Rank	Problem Identification	Operational Impact	Current Client Workaround	Priority Score
1	Total absence of an offline POS system	Forces paper-based order entry; leads to miscalculated totals and lost order slips.	Writing orders on physical notepads.	43
2	Manual post-event sales tallying	Requires 2 to 3 hours of late-night manual calculation per event; causes operator burnout.	Tallying paper receipts using handheld calculators.	42
3	Inefficient product & ingredient inventory tracking	Frequent mid-event stockouts of essential items (e.g., espresso beans, full-cream milk, cold cups).	Visual estimation and post-event manual updates to basic Excel sheets.	38
4	Absence of low-stock alert notifications	Staff remain unaware when raw materials reach critical thresholds, resulting in rejected customer orders.	Manual spot checks during event lulls.	34
5	Lack of historical event performance data	Management cannot identify best-selling products, peak hours, or high-margin events.	Reliance on gut feeling and basic memory.	N/A

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6	Time-consuming administrative reconciliation	High labor fatigue reduces overall operational efficiency and business scalability.	Owners working long unpaid hours after events.	N/A
7	Severe manpower constraints	A small team (3 members) was forced to divide attention between customer service and manual paperwork.	Operations Manager personally handling manual logging during sales.	N/A

Table 1.1. Problem Identification and Priority Ranking

Root-cause stratification demonstrates that operational breakdowns propagate sequentially throughout the business structure. The absence of a dedicated POS system leads directly to manual tracking of orders, which in turn creates a total lack of historical sales data from past events. In parallel, inefficient ingredient tracking causes unexpected stockouts, resulting in unrecorded waste and lost revenue. Furthermore, operating without a permanent brick-and-mortar storefront forces full reliance on catering events, thereby magnifying transportation and operational overhead. When combined with a small operational team handling multiple physical roles, manual administrative tasks severely impair customer service delivery and cause worker burnout.

Background and Rationale

The specialty coffee market in the Philippines has expanded rapidly, driven by rising urban consumption and evolving beverage preferences^[5]. Industry evaluations highlight that domestic coffee demand significantly outpaces local production, fostering vibrant growth in retail coffee service and quick-service mobile setups across Metro Manila and surrounding regions^[5]. A major catalyst of this expansion is the emergence of mobile pop-up coffee vendors—lightweight, highly portable setups operating out of customized motorcycles, pop-up carts, and transportable espresso stations. These micro-enterprises offer low capital expenditure requirements and high location flexibility. However, their mobility creates severe technical constraints: event venues frequently operate in cellular dead zones or congested wireless environments where online point-of-sale software fails entirely.

TravelBean MNL is a mobile pop-up coffee business based in Pampanga St., Tondo, Manila, established in 2024. The venture utilizes a specialized motorcycle setup to serve specialty espresso drinks, cold brew, flavor-infused beverages, and light pastries at catering events, school fairs, weddings, and private parties throughout Metro Manila and nearby provinces^[5].

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The operational structure of TravelBean MNL relies on three primary personnel:

- Rhona Mae Aganan: Owner & Operations Manager, responsible for direct order taking, customer service, administrative reporting, and inventory management.
- Jimwell Mamangun: Co-Owner & Logistics Manager, handling equipment transportation, site setup, stock procurement, and physical logistics.
- Hired Barista: Dedicated staff member focused on beverage preparation, espresso extraction, and order fulfillment.

Pop-up coffee stands present a unique inventory management problem compared to standard retail outlets^[8]. Unlike traditional retail stores that track discrete packaged items (such as selling one canned beverage which deducts exactly one unit from inventory), coffee shops assemble products from multiple raw ingredients^[1]. Preparing a single 16oz Iced Salted Caramel Latte requires the simultaneous consumption of 18 grams of espresso beans, 200 milliliters of full-cream milk, 30 milliliters of salted caramel syrup, one 16oz clear plastic cup, one dome lid, and one plastic straw.

When inventory is tracked only at the finished-product level, management remains blind to actual raw material balances during an event^[1]. VenueVue, developed by group CTRL+Stay, bridges this gap by introducing an offline-first POS engine paired with an automated recipe-mapping database. The system runs on a zero-subscription local client-server architecture, providing immediate operational stability without recurring financial overhead^[1].

Objectives

The general objective of group CTRL+Stay is to design, develop, test, and deploy VenueVue, an offline-first Point-of-Sale and Inventory Management System that automates order processing, sales tracking, ingredient-level inventory deduction, low-stock threshold alerting, and post-event analytical report generation for TravelBean MNL^[1].

#	Specific Objective	Quantifiable Success Metric
1	Digitally capture customer transactions at the point of sale.	100% elimination of paper order slips during event operations.
2	Automate raw material inventory deductions based on product recipe mappings.	Immediate sub-second inventory adjustments per ingredient upon transaction completion ^[4] .
3	Implement real-time low-stock alerting mechanisms.	System displays visual alert prompts whenever ingredient stock falls below predefined thresholds ^[4] .

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4	Guarantee complete offline system availability during mobile events.	100% functional availability with zero internet connectivity required.
5	Automate post-event sales and operational report generation.	Generation of structured 4-tab Excel workbooks (.xlsx) within 10 seconds post-event ^[10] .
6	Enable flexible multi-device data synchronization post-event.	Support for 3 distinct data transfer protocols: Local Wi-Fi push, direct USB transfer, and manual Excel import.
7	Enforce role-based access control (RBAC).	Strict separation of privileges between Baristas (read/order access) and Owners (full administrative control).

Table 1.2. Specific Objectives and Quantifiable Success

Significance and Scope of the Project

VenueVue delivers substantial operational value to all key project stakeholders. For TravelBean MNL Management, the system eliminates post-event calculation errors, cuts administrative reconciliation time by 90%, provides actionable profit analytics, and prevents revenue loss caused by mid-event ingredient stockouts^[4]. For baristas and operations staff, the software streamlines order input through a touch-optimized interface, automates change calculation, eliminates lost order tickets, and allows staff to concentrate on beverage quality and customer engagement. For the broader mobile pop-up industry, VenueVue offers a replicable, cost-effective digital roadmap for micro-vendors operating in constrained offline environments^[6]. Finally, for the academic community, the project serves as a validated software engineering case study combining local client-server network topologies, offline transactional state management, and recipe-mapping relational database schemas.

The system scope encompasses user authentication with role-based access control, a touch-optimized POS ordering interface with category filtering, real-time cart handling, automatic cash change calculation, ingredient-level inventory tracking, a recipe setup engine mapping raw materials to menu items, automated low-stock visual alerts, a multi-method post-event data synchronization module, and automated export of 4-tab Excel reports (Summary, Sales by Product, Ingredient Usage, Restock List)^[4].

To maintain technical feasibility within the project schedule, explicit operational boundaries have been established. Physical receipt printing is intentionally excluded because TravelBean MNL event customers do not request physical paper receipts. Online cloud synchronization is excluded to eliminate internet dependencies during live operations. Hardware barcode scanning is excluded because coffee menu items are selected directly via touchscreen UI grid buttons. Customer loyalty programs, integrated e-wallet payment processing, and

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multi-venue simultaneous site management are likewise excluded as the client operates exclusively on a cash-only, single-event setup basis.

Project Organization

External Interfaces

VenueVue interfaces cleanly with hardware devices, peripheral storage media, and external data formats without relying on external cloud application programming interfaces.

The primary client interaction takes place through the Tablet POS Touch Interface, rendered as a responsive web application inside standard web browsers (Google Chrome or Apple Safari) on 7-inch to 10-inch mobile tablets. Management and administrative functions are handled via the Laptop Administrative Suite, which provides full screen controls on the host laptop for menu configuration, ingredient restocking, recipe mapping, and financial reporting.

Networking across hardware devices is managed via the Local Wi-Fi Network Interface. The host laptop activates its built-in Windows Mobile Hotspot to broadcast a local wireless network operating under a star topology within the 192.168.137.x IP address range. This wireless network allows tablet devices to connect directly to the laptop's local server without requiring external routers or active internet bandwidth.

Physical data transfer and system persistence are supported through the USB Storage Interface, utilizing external USB flash drives for post-event database backup storage (.sql files) and manual order file imports. Lastly, reporting outputs are formatted via the Excel File Generator interface, which leverages the open-source PhpSpreadsheet library to compile database query results into styled .xlsx workbooks^[10].

Internal Structure

The internal system architecture designed by group CTRL+Stay follows a modified Three-Tier Client-Server Pattern deployed strictly within a local, self-contained offline environment^[2].

Architecture Tier	Operational Component & Host Device	Core Technical Logic & Protocols
Client Tier (Presentation Layer)	Mobile Tablets (Android / iPad OS execution via web browser)	Renders touch-optimized POS UI using HTML5, CSS3, Tailwind CSS, and JavaScript. Executes client-side order validation, computes cart totals, and manages temporary local browser session storage.

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Communication Layer		Local Wi-Fi Wireless Hotspot (192.168.137.x subnet)	Handles direct client-server HTTP requests and JSON transaction payloads over local wireless hotspot connection without internet access.
Application (Business Layer)	Tier Logic	Laptop Workstation (Local XAMPP Server Environment)	Executes back-end business logic in PHP 8.x. Processes incoming tablet orders, executes recipe deduction algorithms, enforces RBAC, handles post-event sync, and compiles Excel exports via PhpSpreadsheet ^[10] .
Data (Persistence Layer)	Tier	Laptop Workstation (MySQL 8.x Relational Database Engine)	Stores core business entities across 6 relational tables using prepared statements. Guarantees relational integrity, foreign key constraints, inventory tracking, and audit logging.

Table 2.1. VenueVue Three-Tier System Architecture

The Client Tier executes directly within the web browser of barista tablets. Constructed using HTML5, CSS3, Tailwind CSS, and standard JavaScript, this layer renders the touch-friendly point-of-sale layout, executes client-side order validation, calculates cart totals in real time, and temporarily retains transaction records during offline operating states.

The Application Tier is hosted locally on the laptop workstation using an offline XAMPP web server stack. Executed in PHP 8.x, this layer processes order payloads received from tablets, executes the recipe deduction algorithms to adjust raw ingredient inventories, enforces role-based access security, handles post-event data reconciliation, and compiles multi-tab financial reports.

The Data Tier consists of a MySQL 8.x relational database running locally on the laptop host. This tier guarantees relational integrity, executes foreign key constraints, preserves transaction audit logs, and stores core business data including master ingredient inventories, recipe mappings, menu items, sales records, and hashed user credentials.

Roles and Responsibilities

The software engineering team CTRL+Stay consists of three undergraduate developers from Cavite State University:

Personnel	Academic / Project Role	Primary Project Responsibilities
Zapico, Zaki	Project Manager / Lead Developer	Overall project coordination, timeline tracking, client requirements alignment, SWOT

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		prioritization, risk management, and back-end logic development.
Gregorio, Rhea Mae	System Analyst & Quality Assurance Lead	Requirements gathering, use case modeling, ERD schema design, System Architecture modeling, technical documentation, test strategy creation, and QA oversight.
Roquid, Jenny Lyn	Quality Assurance & Testing Specialist	Test script execution, unit/integration testing, offline sync validation, bug logging, usability assessment, and User Acceptance Testing (UAT) verification.

Table 2.2. Project Roles and Responsibilities

The distribution of responsibilities across the 5-month software development lifecycle is governed by the RACI Framework^[3]:

Project Lifecycle Phase	Zapico, Zaki (PM)	Gregorio, Rhea Mae (SA & QA Lead)	Roquid, Jenny Lyn (QA Specialist)	Client (TravelBean MNL)
Requirement Analysis & SWOT	Accountable / Lead	Responsible	Consulted	Informed / Approver
System Architecture & ERD	Consulted	Accountable / Lead	Consulted	Informed
UI/UX Figma Wireframing	Responsible	Accountable / Lead	Consulted	Approver
Front-End & Back-End Coding	Accountable / Lead	Responsible	Testing	Informed
System Integration & Testing	Responsible	Accountable / Lead	Responsible	Consulted
UAT & Deployment	Accountable / Lead	Support	Responsible	Approver / Sign-Off

Table 2.3. RACI Responsibility Assignment Matrix

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Deliverables

Group CTRL+Stay shall deliver the following formal documentation and software artifacts upon project completion:

Deliverable Item	Format / Medium	Description / Specification
Project Plan Document	PDF / Word (.docx)	Comprehensive operational and managerial project plan.
SWOT Analysis Report	PDF / Word (.docx)	Prioritized SWOT matrix, fishbone diagram, and problem ranking.
Project Technical Doc (PTD)	PDF / Word (.docx)	Detailed technical specifications, network diagrams, and database schemas.
System Architecture Diagram	Vector Graphic / PDF	Full visual mapping of network topologies and hardware components.
Database Design (ERD)	SQL DDL / PDF	Entity-Relationship Diagram with complete attribute data types.
Wireframes / Mockups	Figma Link / PDF	Interactive wireframes covering 7 core system screen layouts.
VenueVue Software System	PHP / JS / SQL Source	Fully functional source code deployed within XAMPP environment.
User Operational Manual	PDF Document	Concise operational guide for baristas and business owners.
System Installation Guide	PDF Document	Step-by-step setup guide covering XAMPP, MySQL, and hotspot setup.
Excel Export Templates	.xlsx Files	Pre-formatted, 4-tab structured financial and stock report layouts ^[10] .
Test Plan & Bug Logs	PDF / Word (.docx)	Completed test scripts, execution logs, and UAT sign-off documentation.

Table 2.4. Project Deliverables and Specifications

Review of Existing Alternatives

To evaluate commercial POS software options, group CTRL+Stay analyzed three relevant market alternatives: Lavu POS^[1], digabloPos^[14], and Peddlr POS^[16]. The evaluation focused on offline stability, ingredient-level tracking capabilities, recipe modeling, low-stock threshold alerts, exportable report formats, and total cost of ownership.

System Feature / Criterion	Lavu POS	digabloPos	Peddlr POS	VenueVue (By CTRL+Stay)
Full Offline Operation	Partial (Limited local caching)	Yes (100% offline-first architecture) ^[14]	Yes (Offline mode supported) ^[16]	Yes (100% offline via local XAMPP server)
Ingredient-Level Tracking	Yes (Monitors raw materials) ^[1]	No (Tracks products & batches only) ^[14]	No (Tracks discrete product units) ^[16]	Yes (Tracks raw inputs in grams, mL, units)
Recipe-Based Auto Deductions	No (Requires cloud add-ons)	No (Manual inventory adjustments) ^[14]	No (Product-level inventory only) ^[16]	Yes (Automated multi-ingredient deduction)
Low-Stock Alerting	Yes (Visual notifications) ^[8]	Yes (Low-stock system notifications) ^[14]	No (Requires manual stock checks) ^[16]	Yes (Configurable threshold warnings)
Exportable Excel Reports	Yes (Standard financial exports)	No (Basic native app analytics) ^[14]	No (Basic sales/credit ledgers) ^[16]	Yes (Tailored 4-tab .xlsx workbooks) ^[10]
Pricing & Licensing Model	Monthly Subscription (\$69+/mo) ^[1]	Free for life / Cloud sync option ^[14]	Free Mobile App ^[16]	One-time free local deployment

Table 3.1. Comparison of Existing POS Alternatives

While Lavu POS provides restaurant-grade inventory tracking, its offline operating mode is secondary to its cloud architecture. It relies on persistent internet connectivity to synchronize complex database entities, and its monthly subscription fees create an unsustainable financial burden for micro pop-up vendors.

digabloPos is a 100% offline-first application supporting multi-currency transactions, table management, and retail tracking^[14]. However, its inventory engine focuses on pre-packaged retail SKUs rather than assembled recipes. It cannot perform multi-ingredient deductions (such as automatically reducing espresso

beans, fresh milk, and paper cups simultaneously upon selling a coffee drink) and lacks custom four-tab Excel report generation tailored for event management^[10].

Peddlr POS is a popular mobile application in the Philippines designed for retail sari-sari stores^[16]. It handles basic product sales, expense ledgers, and e-loading offline^[16]. However, Peddlr tracks inventory exclusively at the item level^[16]. It lacks recipe mapping, cannot perform ingredient-level tracking, provides no real-time stock alerts during high-volume event rushes, and cannot export granular raw material consumption reports^[16].

VenueVue directly resolves these market gaps. Operating on a local XAMPP host via a dedicated Wi-Fi hotspot, VenueVue guarantees 100% operational uptime without internet connectivity. It automatically deducts raw materials per sale based on custom recipe maps, alerts baristas to critical stock thresholds, exports comprehensive four-tab Excel workbooks, and imposes zero subscription costs.

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Approach to be taken in this subject

Theoretical Framework / Methods to be used

The development methodology executed by group CTRL+Stay synthesizes four established software engineering frameworks: the Systems Development Life Cycle (SDLC Waterfall), Agile Scrum Development, User-Centered Design (UCD), and the ISO/IEC 25010 Software Quality Model.

The classic SDLC Waterfall framework governs overall documentation structure, architectural planning, and project milestone verification, ensuring strict compliance with academic requirements. During active coding, Agile Scrum practices are employed in two-week sprint cycles over the 5-month duration. This allows the team to rapidly prototype touch layouts and incorporate feedback from TravelBean MNL owners regarding order entry speed and cart management.

User-Centered Design (UCD) guides interface creation to ensure the point-of-sale screens remain intuitive for baristas working under fast-paced event conditions^[7]. Touch targets are oversized, menu navigation relies on simple category filtering, and checkout interactions are streamlined to minimize taps.

System performance is bench marked against the ISO/IEC 25010 Quality Model, focusing on operational reliability (guaranteeing fault-free offline execution), functional suitability (ensuring accurate recipe inventory deductions), usability (maximizing touch interface speed), and security (enforcing password hashing and input sanitization)^[4].

Rationale for the Framework / Methods

Combining SDLC and Agile methodology balances formal academic governance with practical software flexibility. The upfront structured design required by SDLC guarantees comprehensive technical planning and database schema validation. Simultaneously, Agile sprint iterations allow the development team to adjust user interface elements based on direct feedback from client baristas.

Applying User-Centered Design is critical because baristas operate in high-pressure event environments where complex or slow software interfaces cause service delays. Focusing on UCD ensures the system can be mastered by new staff within 15 minutes of training.

Adopting ISO/IEC 25010 metrics establishes objective technical performance targets, ensuring system stability during live event operations.

Technologies you plan to consider

System Layer	Selected Technology	Version	Technical Rationale for Choice
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Front-End UI	HTML5 / CSS3 / JavaScript	ES6+	Native browser execution on any tablet without requiring compiled mobile applications.
CSS Framework	Tailwind CSS	v3.x	Lightweight, responsive utility classes supporting touch-optimized button layouts.
Back-End Engine	PHP	v8.x	Native server-side script execution within XAMPP; broad offline local environment compatibility.
Database Engine	MySQL	v8.x	Robust relational database supporting complex JOIN queries between recipes and stock.
Local Server Host	XAMPP Stack	v8.2+	Free, cross-platform local Apache server bundle featuring offline MySQL execution.
Excel Generator	PhpSpreadsheet	v1.28+	Open-source PHP library capable of generating styled multi-tab .xlsx files locally ^[10] .
Design Wireframes	Figma	Desktop	Rapid visual prototyping of touch-friendly tablet screen layouts.
Version Control	Git / GitHub	v2.x	Distributed source code repository management and team code collaboration.

Table 4.1. Selected Technologies

System Design

Requirements

The system must satisfy explicit functional and non-functional engineering requirements:

The User Authentication module (FR-01) must enforce role-based login requiring valid credentials, directing users to barista or owner interfaces. The POS Touch Order Entry module (FR-02) must render products in category tabs and permit single-tap item selection. The Payment module (FR-03) must accept cash inputs and automatically compute change due. The Inventory Engine (FR-04) must deduct raw ingredients automatically based on mapped product recipes upon order completion. The Alerting Engine (FR-05) must display visual low-stock warnings when inventory drops below predefined thresholds. The Data Sync module (FR-06) must support order ingestion via local Wi-Fi, USB file copy, or Excel import. The Reporting module (FR-07) must export 4-tab Excel workbooks (Summary, Sales by Product, Ingredient Usage, Restock List).

Non-functional requirements demand 100% offline availability without active WAN connections (NFR-01), UI touch response times under 200 milliseconds (NFR-02), secure password hashing via `password_hash()` and SQL injection prevention via Prepared Statements (NFR-03)^[9], and daily automated .sql database backups (NFR-04).

- Client-side transaction validation rules prevent data corruption during event sales:
- Empty orders containing zero line items are rejected.
- Item quantities must be positive integers greater than zero.
- Unit prices are locked to internal system menu tables and cannot be manually overridden by baristas.
- Each order receives a unique identifier formatted as [TabletID]-[SequentialNo] (e.g., T1-001)³.

Offline error handling and conflict resolution procedures maintain data integrity across operating states:

Error Scenario	Technical Detection	Automated System Action	Operational Recovery Procedure
Tablet POS App Crash	Browser session drop.	Background auto-save preserves order cache in local browser memory every 5 orders.	Barista reopens browser; pending order cache is restored automatically.

Wi-Fi Sync Network Drop	Connection timeout after 3 retries.	Local sync process pauses; order records remain cached locally on tablet ^[1] .	Owner switches sync mode to USB transfer (Method 2) or Excel import (Method 3).
Duplicate Order Submission	Primary key on collision on Order_ID.	Laptop import engine flags duplicate ID and pauses auto-merge.	Owner reviews timestamps on laptop, selects valid record, and discards duplicate.
File Import Corruption	PHP file parsing error.	Laptop displays "Import Failed" dialog; database state remains unchanged.	Owner inspects source .xlsx/.csv file, fixes formatting, or re-exports file from tablet ^[10] .

Table 5.1. Offline Error Handling and Conflict Resolution Procedures

Resources

Hardware assets are divided between the central server laptop and barista client tablets:

Component	Workstation / Laptop (Server Host)	Client Tablet (Barista POS Device)
Device Type	Standard Business Laptop	Android Tablet or Apple iPad
Operating System	Windows 10 or Windows 11 (64-bit)	Android 8.0+ or iPadOS 14+
Processor (CPU)	Intel Core i3 (8th Gen) / AMD Ryzen 3 or higher	Octa-Core 1.6 GHz or higher
System Memory (RAM)	4 GB minimum (8 GB recommended)	2 GB minimum (4 GB recommended)
Internal Storage	128 GB SSD minimum	16 GB internal storage
Display Panel	13.3-inch display (1920x1080)	7.0-inch to 10.5-inch Touchscreen
Wireless Module	Wi-Fi 802.11 b/g/n (Hotspot Capable)	Wi-Fi 802.11 b/g/n
Peripherals	External USB 3.0 Flash Drive (Backup)	Stylus or capacitive touch screen

Table 5.2. Hardware Resource Requirements

The network operates as a local wireless star topology centered on the host laptop. The laptop generates a local Wi-Fi hotspot using Windows Mobile Hotspot.

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The laptop server is statically assigned IP 192.168.137.1, while connecting tablets receive DHCP leases within the 192.168.137.100+ subnet. No external router or active internet connection is required.

System Overview

System responsibilities are partitioned across six functional core modules³:

Core Module	Access Level	Primary Technical Responsibilities
User Authentication	All Users	Validates credentials, sets session security tokens, and routes users to appropriate UI layouts.
POS Order Management	Barista & Owner	Renders touchscreen grid, handles cart adjustments, calculates cash change, and processes checkout.
Inventory Management	Owner Only	Monitors raw ingredient quantities, supports manual stock additions, and manages threshold alerts.
Recipe Management	Owner Only	Maps specific raw ingredient quantities (grams, mL, units) to individual menu products.
Post-Event Data Sync	Owner Only	Manages ingestion of tablet order data via Wi-Fi push, USB import, or Excel file upload.
Reporting & Analytics	Owner Only	Executes multi-table relational SQL queries and compiles 4-tab Excel workbooks (.xlsx).

Table 5.3.VenueVue Functional Modules

During an event, tablets operate completely offline, capturing sales and caching transactions locally. Devices do not communicate with each other or the laptop during active sales hours.

Once the event concludes, data reconciliation occurs using one of three ingestion protocols:

- Method 1 (Wi-Fi Hotspot Sync): The owner enables the laptop's Wi-Fi hotspot. Tablets connect and push stored JSON order records directly to the laptop's PHP backend over local HTTP, populating the MySQL database.
- Method 2 (USB Cable Transfer): The owner connects tablets directly to the laptop via USB cable and copies exported transaction files into the laptop's ingestion directory for processing.
- Method 3 (Manual Excel/CSV Upload): Tablets export sales logs to a USB flash drive as .xlsx or .csv files. The owner inserts the drive into the laptop and imports the files using the administrative upload tool.

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The database layer consists of six relational tables configured in MySQL 8.x3:

1. **USERS:** Stores account credentials, role privileges, and hashed passwords.
2. **PRODUCTS:** Maintains menu item names, categories, and retail prices.
3. **INGREDIENTS:** Tracks raw material stock levels, unit measures (grams, mL, units), and low-stock alert thresholds.
4. **RECIPE_ITEMS:** Junction table linking products to ingredients, defining required raw material quantities per item sold.
5. **ORDERS:** Records transaction totals, cash tendered, and timestamps.
6. **ORDER_ITEMS:** Captures individual line items per order, driving automated recipe inventory deductions.

Project Timeline

Gantt Chart

The software engineering lifecycle spans 5 Months (20 Weeks), divided across five primary phases:

Figure 1.1. 5-Month Execution Schedule

Work Breakdown Structure

VENUEVUE SOFTWARE DEVELOPMENT PROJECT (Group CTRL+Stay)

Phase 1: Project Initiation & Requirements Engineering (Month 1)

- 1.1 Conduct on-site operational study at TravelBean MNL pop-up setup.
- 1.2 Execute SWOT analysis and priority scoring matrix.
- 1.3 Construct Fishbone cause-and-effect problem stratification charts.
- 1.4 Formalize functional and non-functional engineering requirements.
- 1.5 Author and finalize Software Project Plan and Project Charter.

Phase 2: Architectural & System Design (Month 2)

- 2.1 Design relational 6-table MySQL ERD schema and DDL script files.

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- 2.2 Specify local Wi-Fi star network topology (192.168.137.x).
- 2.3 Build interactive Figma touch UI mockups for 7 system screens.
- 2.4 Draft Project Technical Documentation (PTD).
- 2.5 Conduct architectural design review with academic evaluators.

Phase 3: Development & System Coding (Months 3–4)

- 3.1 Configure local XAMPP Apache host and MySQL 8.x database engine.
- 3.2 Program secure User Authentication module with password_hash().
- 3.3 Build touch-optimized POS ordering interface with category filtering.
- 3.4 Develop a back-end PHP engine for recipe-based inventory deduction.
- 3.5 Construct low-stock alert notifications and dashboard warning badges.
- 3.6 Implement a multi-method post-event sync engine (Wi-Fi push, USB, Excel upload).
- 3.7 Integrate PhpSpreadsheet library for automated 4-tab .xlsx report creation.

Phase 4: System Verification & Quality Assurance (Month 5)

- 4.1 Conduct automated unit testing on recipe calculation formulas.
- 4.2 Perform integration testing across host laptop and barista client tablets.
- 4.3 Simulate network disconnections and corrupted file import scenarios.
- 4.4 Log, prioritize, and resolve software defects in issue tracker.

Phase 5: Deployment, Training & Formal Sign-Off (Month 5)

- 5.1 Deploy final VenueVue production software stack onto client laptop.
- 5.2 Conduct hands-on training sessions for TravelBean MNL baristas and owners.
- 5.3 Deliver User Operational Manual and System Installation Guide.
- 5.4 Execute User Acceptance Testing (UAT) scripts.
- 5.5 Secure signed formal acceptance documentation from client and university board.

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Budget Allocation

Estimated Cost

Capital outlays are minimized by utilizing open-source development tools and client-provided hardware assets.

Software licenses are entirely free, as XAMPP, PHP 8.x, MySQL 8.x, Tailwind CSS, PhpSpreadsheet, and Figma operate under open-source or free usage tiers. Direct hardware expenses total ₱900.00 for two high-speed 64GB USB flash drives used for secondary offline database backups. The server workstation (valued at ₱25,000.00) and barista tablets (valued at ₱8,000.00) are supplied directly by TravelBean MNL.

Academic labor valuations over the 5-month timeline reflect 650 total development hours for group CTRL+Stay:

- Project Manager & Lead Developer Zapico (250 hrs @ ₱250/hr = ₱62,500.00)
- System Analyst & QA Lead Gregorio (220 hrs @ ₱200/hr = ₱44,000.00)
- Quality Assurance Specialist Roquid (180 hrs @ ₱180/hr = ₱32,400.00)
- Total Imputed Team Labor Valuation: ₱138,900.00

Direct operational expenses over the 5-month duration total ₱10,700.00, covering client site visitation travel to Tondo, Manila (₱4,000.00), documentation printing and binding (₱2,200.00), deployment testing supplies (₱2,500.00), and a project contingency buffer (₱2,000.00).

Total Project Budget

Budget Allocation Category	Direct Out-of-Pocket Cost (PHP)	Imputed Project Valuation (PHP)
Software Development & Licenses	₱0.00	₱0.00
Hardware & Storage Media	₱900.00	₱33,900.00
Software Engineering Labor (CTRL+Stay)	₱0.00	₱138,900.00
Operational & Deployment Outlays	₱10,700.00	₱10,700.00
TOTAL PROJECTED EXPENDITURE	₱11,600.00	₱183,500.00

Table 6.1. Project Budget Allocation

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The total direct out-of-pocket expenditure required over the 5-month project lifecycle is ₱11,600.00. By leveraging open-source software frameworks and reallocating existing hardware, group CTRL+Stay delivers a commercial-grade offline POS system valued at ₱183,500.001.

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